Nhi Dinh

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EDUCATION

University of Illinois Chicago — Chicago, IL
Ph.D. in Mathematics (Mathematical Computer Science) — Aug 2026 - Present
M.S. in Mathematical Computer Science — Aug 2026

University of Illinois Urbana-Champaign — Champaign, IL
B.S. in Computer Science, Statistics, Mathematics; GPA: 3.9/4.0 — May 2026
Honors: Highest Distinction; Fiddler Innovation Fellowship; x4 Dean's List
Coursework: Machine Learning, Deep Learning, Algorithms, Probability Theory, Applied Bayesian Analysis
Additional Training: Erdős Institute - Deep Learning Boot Camp

SKILLS

Languages: Python, SQL, C++, C, R | **ML/NLP:** Sentence Transformers, PyTorch, scikit-learn, GraphRAG
Methods: Embeddings, cosine retrieval, clustering, SVD, random projection, benchmark design, error analysis
Tools: NumPy, pandas, SciPy, FastAPI, Streamlit, LLM APIs, Git, Linux, Docker, pytest, GitLab CI/CD

PROJECTS

NewsTrace - NLP Retrieval & Streaming News Intelligence
Python, Sentence Transformers, PyTorch, scikit-learn, FastAPI
- Built a streaming-news NLP system with sentence-transformer embeddings, cosine retrieval, and online clustering; implemented PyTorch device handling.
- Designed reproducible experiments comparing full-dimensional embeddings with SVD and random-projection compression using retrieval, clustering, latency, and memory metrics.
- Developed evidence-grounded retrieval and summarization workflows with deduplication, provenance tracking, and FastAPI/Streamlit interfaces.

Vietnamese GraphRAG Question Answering - AICustomerBot
Python, Microsoft GraphRAG, Embeddings, LLM APIs
- Integrated Microsoft's GraphRAG into Vietnamese document QA using entity/relationship extraction, knowledge graphs, embeddings, and local/global semantic search.
- Extended the workflow with source excerpts, search fallback, prompt configuration, response formatting, and evaluation utilities for grounded QA.

SpecShield - LLM Agent Safety & Evaluation — *Python, LLM APIs, AgentDojo*
- Developed a runtime monitor for tool-using LLM agents with typed action fields, information-flow tracking, approval provenance, and deterministic authorization policies.
- Built evaluation harnesses for prompt injection, benign-task false blocks, adversarial behavior, provenance corruption, and multi-step trajectories across four AgentDojo suites and six bilingual scenarios.

EXPERIENCE

MB Securities JSC — Hanoi, Vietnam
Data Scientist Intern — Jun 2024 - Aug 2024
- Built and evaluated Vietnamese/English financial document QA using GraphRAG-based retrieval and grounded citations.
- Constructed a 200-query labeled benchmark and performed error analysis; improved citation precision by 63% and reduced unsupported claims by 50%.

Square Kettle — Illinois, USA
Quantitative Research Intern — Jan 2026 - July 2026
- Built a reproducible Python pipeline for 200+ intraday backtests with two-stage Optuna/TPE tuning over 13 parameters, walk-forward validation, risk constraints, automated metrics, Docker, and AWS S3 artifact tracking.
- Improved out-of-sample backtested Calmar from 1.62 to 3.33 and return from 7.6% to 15.4%, with maximum drawdown near 4.6%, by tuning signals and ATR-based exits on 1-minute Nasdaq-100 E-mini futures.
- Developed graph-Laplacian and discrete-wave models of cross-asset price and volatility shocks; benchmarked against diffusion, mean-reversion, and statistical baselines on unseen periods.

Congero Technology — Virginia, USA
Data Developer Intern — Jun 2025 - Aug 2025
- Implemented LLM API onboarding, token rating, and balance tracking with Oracle BRM C opcodes across 12 service plans at 10K+ transactions/hour.
- Built event schemas, data-quality checks, batch loading, and Docker/GitLab CI/CD; reduced deployment time to under five minutes for a July 2025 production launch.

PUBLICATIONS

Algorithmic Phase Transition for Large Independent Sets in Dense Hypergraphs — RANDOM 2026
Choosability of Multipartite Hypergraphs — Discrete Mathematics